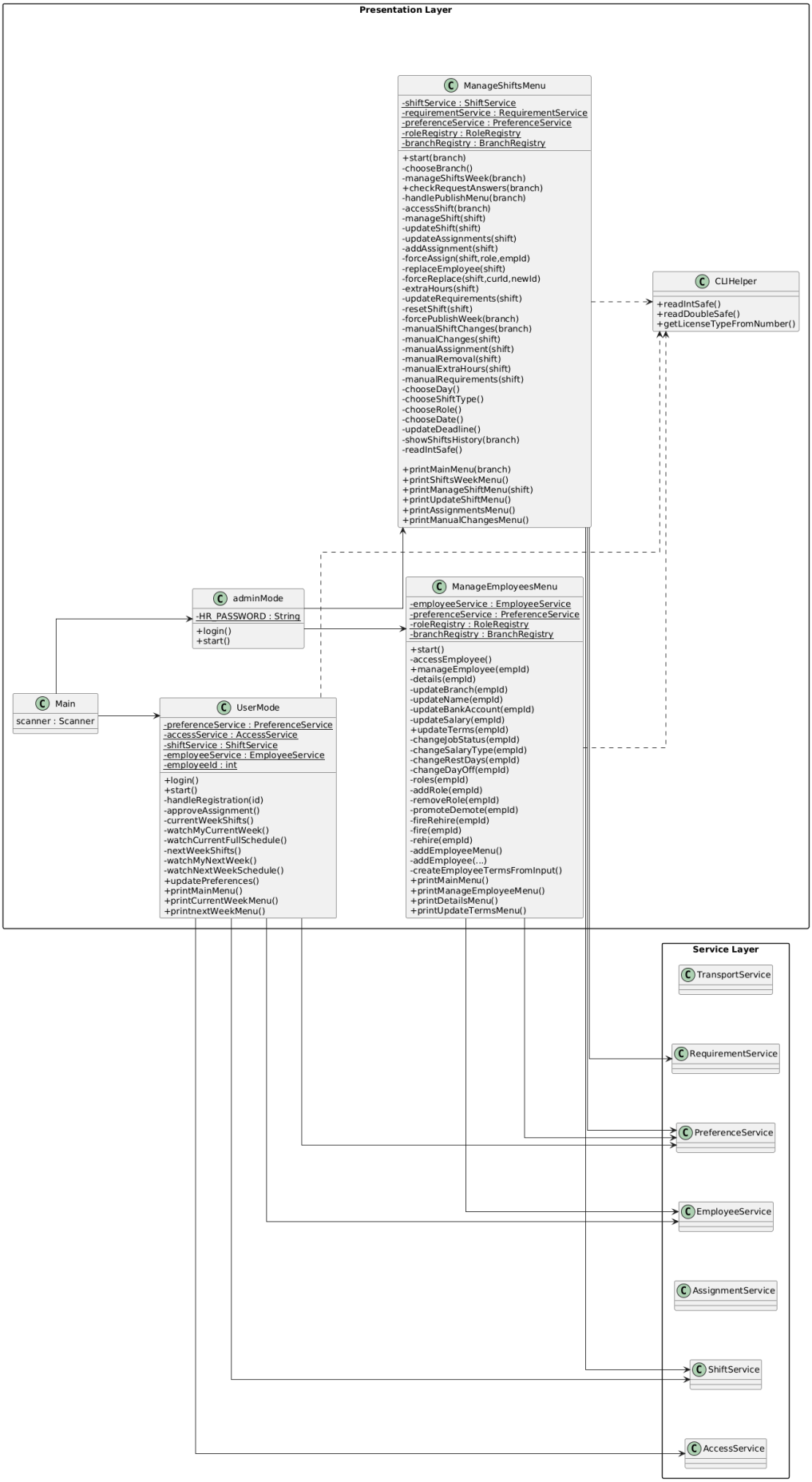
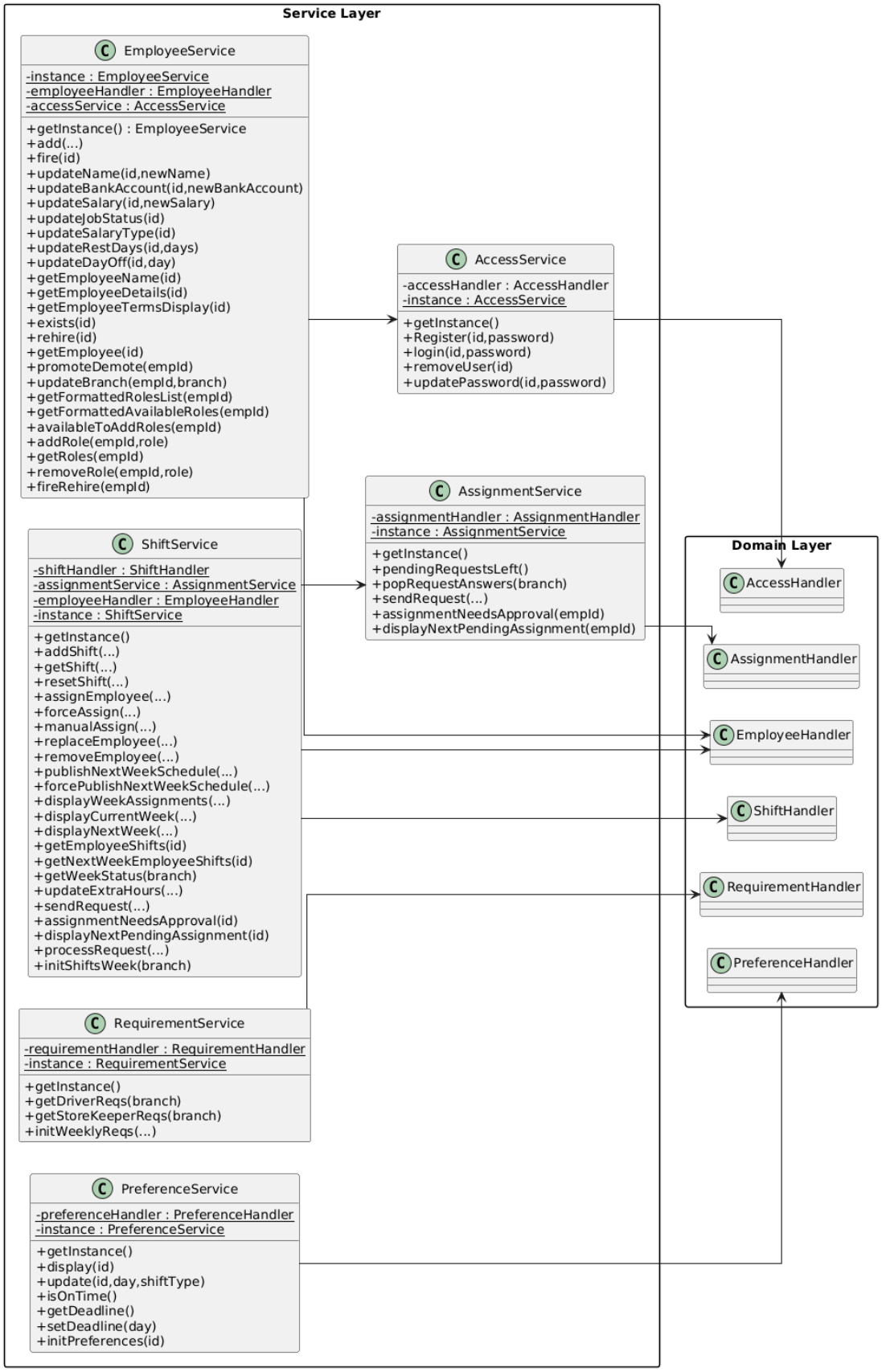
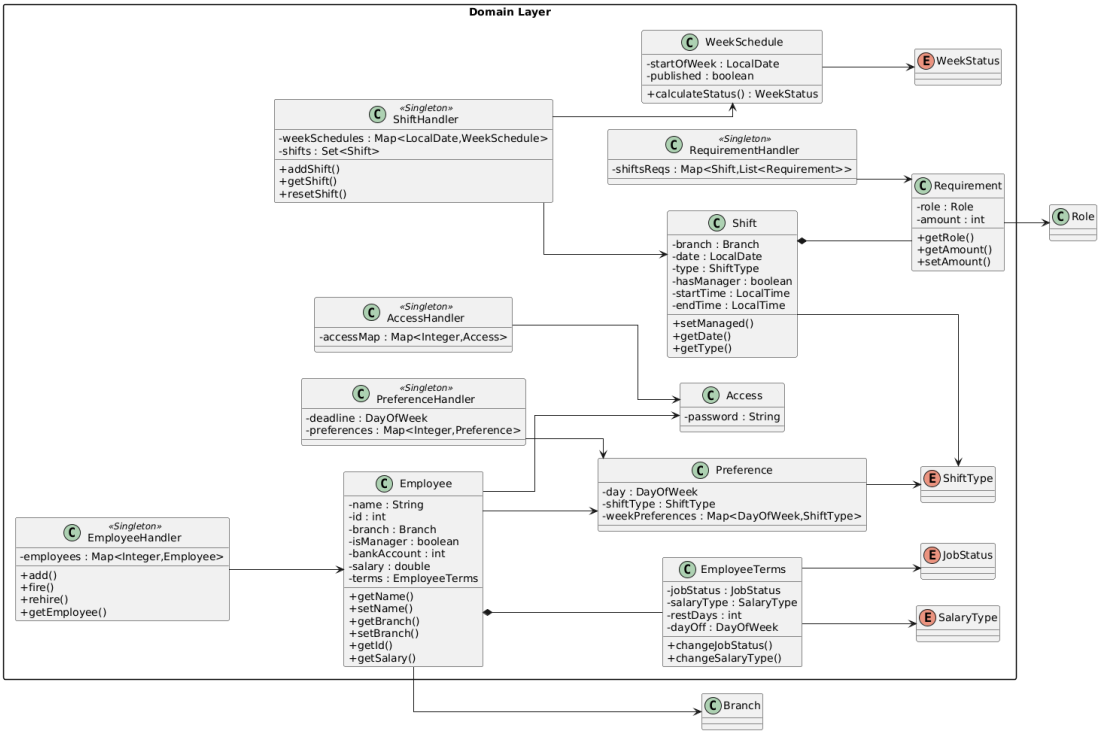
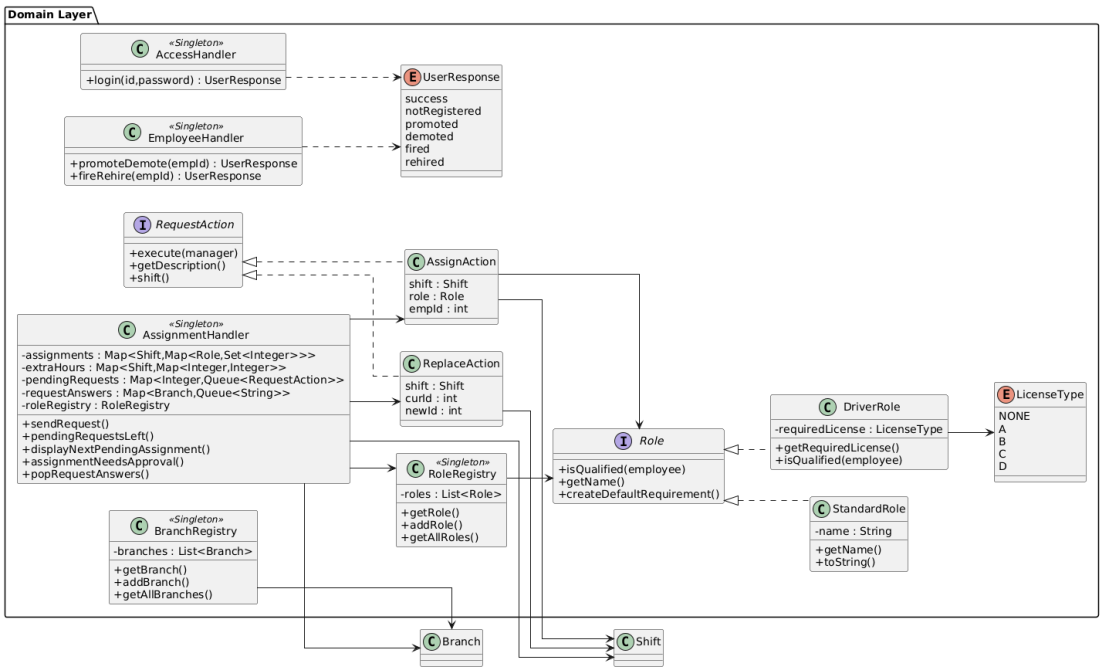
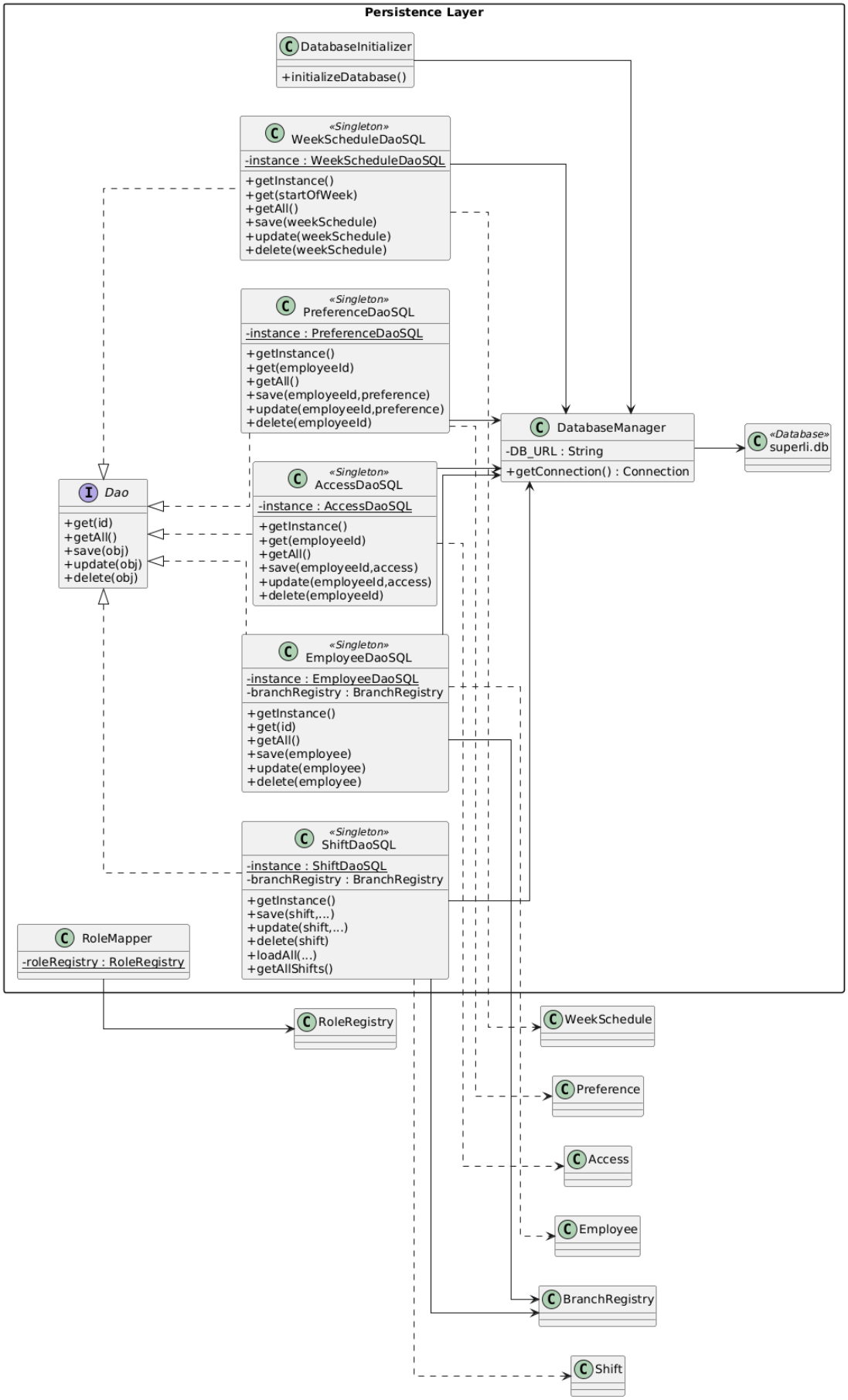
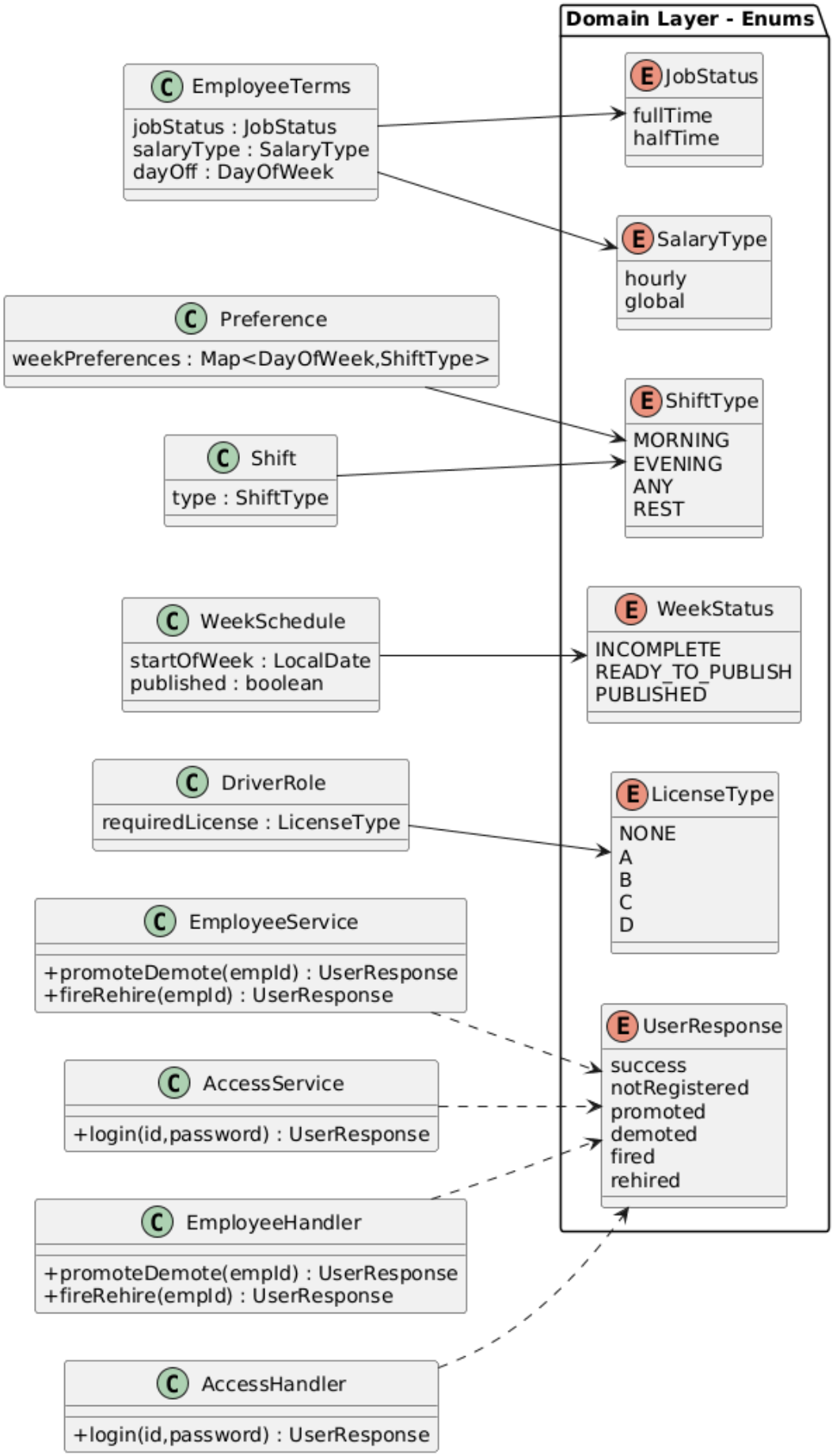

Presentation Layer
ManageShiftsMenu
-shiftService : ShiftService
-requirementService : RequirementService
-preferenceService : PreferenceService
-roleRegistry : RoleRegistry
-branchRegistry : BranchRegistry
+start(branch)
-chooseBranch()
-manageShiftsWeek(branch)
+checkRequestAnswers(branch)
-handlePublishMenu(branch)
-accessShift(branch)
-manageShift(shift)
-updateShift(shift)
-updateAssignments(shift)
-addAssignment(shift)
-forceAssign(shift,role,empId)
-replaceEmployee(shift)
-forceReplace(shift,curId,newId)
-extraHours(shift)
-updateRequirements(shift)
-resetShift(shift)
-forcePublishWeek(branch)
-manualShiftChanges(branch)
-manualChanges(shift)
-manualAssignment(shift)
-manualRemoval(shift)
-manualExtraHours(shift)
-manualRequirements(shift)
-chooseDay()
-chooseShiftType()
-chooseRole()
-chooseDate()
-updateDeadline()
-showShiftsHistory(branch)
-readIntSafe()
+printMainMenu(branch)
+printShiftsWeekMenu()
+printManageShiftMenu(shift)
+printUpdateShiftMenu()
+printAssignmentsMenu()
+printManualChangesMenu()
CLIHelper
+readIntSafe()
+readDoubleSafe()
+getLicenseTypeFromNumber()
adminMode
-HR_PASSWORD : String
+login()
+start()
ManageEmployeesMenu
-employeeService : EmployeeService
-preferenceService : PreferenceService
-roleRegistry : RoleRegistry
-branchRegistry : BranchRegistry
+start()
-accessEmployee()
+manageEmployee(empId)
-details(empId)
-updateBranch(empId)
-updateName(empId)
-updateBankAccount(empId)
-updateSalary(empId)
+updateTerms(empId)
-changeJobStatus(empId)
-changeSalaryType(empId)
-changeRestDays(empId)
-changeDayOff(empId)
-roles(empId)
-addRole(empId)
-removeRole(empId)
-promoteDemote(empId)
-fireRehire(empId)
-fire(empId)
-rehire(empId)
-addEmployeeMenu()
-addEmployee(...)
-createEmployeeTermsFromInput()
+printMainMenu()
+printManageEmployeeMenu()
+printDetailsMenu()
+printUpdateTermsMenu()
Main
scanner : Scanner
UserMode
-preferenceService : PreferenceService
-accessService : AccessService
-shiftService : ShiftService
-employeeService : EmployeeService
-employeeId : int
+login()
+start()
-handleRegistration(id)
-approveAssignment()
-currentWeekShifts()
-watchMyCurrentWeek()
-watchCurrentFullSchedule()
-nextWeekShifts()
-watchMyNextWeek()
-watchNextWeekSchedule()
+updatePreferences()
+printMainMenu()
+printCurrentWeekMenu()
+printnextWeekMenu()
Service Layer
TransportService
RequirementService
PreferenceService
EmployeeService
AssignmentService
ShiftService
AccessService


Service Layer
EmployeeService
-instance : EmployeeService
-employeeHandler : EmployeeHandler
-accessService : AccessService
+getInstance() : EmployeeService
+add(...)
+fire(id)
+updateName(id,newName)
+updateBankAccount(id,newBankAccount)
+updateSalary(id,newSalary)
+updateJobStatus(id)
+updateSalaryType(id)
+updateRestDays(id,days)
+updateDayOff(id,day)
+getEmployeeName(id)
+getEmployeeDetails(id)
+getEmployeeTermsDisplay(id)
+exists(id)
+rehire(id)
+getEmployee(id)
+promoteDemote(empId)
+updateBranch(empId,branch)
+getFormattedRolesList(empId)
+getFormattedAvailableRoles(empId)
+availableToAddRoles(empId)
+addRole(empId,role)
+getRoles(empId)
+removeRole(empId,role)
+fireRehire(empId)
AccessService
-accessHandler : AccessHandler
-instance : AccessService
+getInstance()
+Register(id,password)
+login(id,password)
+removeUser(id)
+updatePassword(id,password)
AssignmentService
-assignmentHandler : AssignmentHandler
-instance : AssignmentService
+getInstance()
+pendingRequestsLeft()
+popRequestAnswers(branch)
+sendRequest(...)
+assignmentNeedsApproval(empId)
+displayNextPendingAssignment(empId)
ShiftService
-shiftHandler : ShiftHandler
-assignmentService : AssignmentService
-employeeHandler : EmployeeHandler
-instance : ShiftService
+getInstance()
+addShift(...)
+getShift(...)
+resetShift(...)
+assignEmployee(...)
+forceAssign(...)
+manualAssign(...)
+replaceEmployee(...)
+removeEmployee(...)
+publishNextWeekSchedule(...)
+forcePublishNextWeekSchedule(...)
+displayWeekAssignments(...)
+displayCurrentWeek(...)
+displayNextWeek(...)
+getEmployeeShifts(id)
+getNextWeekEmployeeShifts(id)
+getWeekStatus(branch)
+updateExtraHours(...)
+sendRequest(...)
+assignmentNeedsApproval(id)
+displayNextPendingAssignment(id)
+processRequest(...)
+initShiftsWeek(branch)
RequirementService
-requirementHandler : RequirementHandler
-instance : RequirementService
+getInstance()
+getDriverReqs(branch)
+getStoreKeeperReqs(branch)
+initWeeklyReqs(...)
PreferenceService
-preferenceHandler : PreferenceHandler
-instance : PreferenceService
+getInstance()
+display(id)
+update(id,day,shiftType)
+isOnTime()
+getDeadline()
+setDeadline(day)
+initPreferences(id)
Domain Layer
AccessHandler
AssignmentHandler
EmployeeHandler
ShiftHandler
RequirementHandler
PreferenceHandler


Domain Layer
WeekSchedule
-startOfWeek : LocalDate
-published : boolean
+calculateStatus() : WeekStatus
WeekStatus
«Singleton»
ShiftHandler
-weekSchedules : Map<LocalDate,WeekSchedule>
-shifts : Set<Shift>
+addShift()
+getShift()
+resetShift()
«Singleton»
RequirementHandler
-shiftsReqs : Map<Shift,List<Requirement>>
Requirement
-role : Role
-amount : int
+getRole()
+getAmount()
+setAmount()
Role
Shift
-branch : Branch
-date : LocalDate
-type : ShiftType
-hasManager : boolean
-startTime : LocalTime
-endTime : LocalTime
+setManaged()
+getDate()
+getType()
«Singleton»
AccessHandler
-accessMap : Map<Integer,Access>
«Singleton»
PreferenceHandler
-deadline : DayOfWeek
-preferences : Map<Integer,Preference>
Access
-password : String
Preference
-day : DayOfWeek
-shiftType : ShiftType
-weekPreferences : Map<DayOfWeek,ShiftType>
ShiftType
Employee
-name : String
-id : int
-branch : Branch
-isManager : boolean
-bankAccount : int
-salary : double
-terms : EmployeeTerms
+getName()
+setName()
+getBranch()
+setBranch()
+getId()
+getSalary()
«Singleton»
EmployeeHandler
-employees : Map<Integer,Employee>
+add()
+fire()
+rehire()
+getEmployee()
EmployeeTerms
-jobStatus : JobStatus
-salaryType : SalaryType
-restDays : int
-dayOff : DayOfWeek
+changeJobStatus()
+changeSalaryType()
JobStatus
SalaryType
Branch


Domain Layer
«Singleton»
AccessHandler
+login(id,password) : UserResponse
UserResponse
success
notRegistered
promoted
demoted
fired
rehired
«Singleton»
EmployeeHandler
+promoteDemote(empId) : UserResponse
+fireRehire(empId) : UserResponse
RequestAction
+execute(manager)
+getDescription()
+shift()
AssignAction
shift : Shift
role : Role
empId : int
«Singleton»
AssignmentHandler
-assignments : Map<Shift,Map<Role,Set<Integer>>>
-extraHours : Map<Shift,Map<Integer,Integer>>
-pendingRequests : Map<Integer,Queue<RequestAction>>
-requestAnswers : Map<Branch,Queue<String>>
-roleRegistry : RoleRegistry
+sendRequest()
+pendingRequestsLeft()
+displayNextPendingAssignment()
+assignmentNeedsApproval()
+popRequestAnswers()
ReplaceAction
shift : Shift
curId : int
newId : int
Role
+isQualified(employee)
+getName()
+createDefaultRequirement()
DriverRole
-requiredLicense : LicenseType
+getRequiredLicense()
+isQualified(employee)
LicenseType
NONE
A
B
C
D
«Singleton»
RoleRegistry
-roles : List<Role>
+getRole()
+addRole()
+getAllRoles()
StandardRole
-name : String
+getName()
+toString()
«Singleton»
BranchRegistry
-branches : List<Branch>
+getBranch()
+addBranch()
+getAllBranches()
Branch
Shift


Persistence Layer
DatabaseInitializer
+initializeDatabase()
«Singleton»
WeekScheduleDaoSQL
-instance : WeekScheduleDaoSQL
+getInstance()
+get(startOfWeek)
+getAll()
+save(weekSchedule)
+update(weekSchedule)
+delete(weekSchedule)
«Singleton»
PreferenceDaoSQL
-instance : PreferenceDaoSQL
+getInstance()
+get(employeeId)
+getAll()
+save(employeeId,preference)
+update(employeeId,preference)
+delete(employeeId)
DatabaseManager
-DB_URL : String
+getConnection() : Connection
«Database»
superli.db
Dao
+get(id)
+getAll()
+save(obj)
+update(obj)
+delete(obj)
«Singleton»
AccessDaoSQL
-instance : AccessDaoSQL
+getInstance()
+get(employeeId)
+getAll()
+save(employeeId,access)
+update(employeeId,access)
+delete(employeeId)
«Singleton»
EmployeeDaoSQL
-instance : EmployeeDaoSQL
-branchRegistry : BranchRegistry
+getInstance()
+get(id)
+getAll()
+save(employee)
+update(employee)
+delete(employee)
«Singleton»
ShiftDaoSQL
-instance : ShiftDaoSQL
-branchRegistry : BranchRegistry
+getInstance()
+save(shift,...)
+update(shift,...)
+delete(shift)
+loadAll(...)
+getAllShifts()
RoleMapper
-roleRegistry : RoleRegistry
RoleRegistry
WeekSchedule
Preference
Access
Employee
BranchRegistry
Shift


Domain Layer - Enums
EmployeeTerms
jobStatus : JobStatus
salaryType : SalaryType
dayOff : DayOfWeek
JobStatus
fullTime
halfTime
SalaryType
hourly
global
Preference
weekPreferences : Map<DayOfWeek,ShiftType>
ShiftType
MORNING
EVENING
ANY
REST
Shift
type : ShiftType
WeekSchedule
startOfWeek : LocalDate
published : boolean
WeekStatus
INCOMPLETE
READY_TO_PUBLISH
PUBLISHED
DriverRole
requiredLicense : LicenseType
LicenseType
NONE
A
B
C
D
EmployeeService
+promoteDemote(empId) : UserResponse
+fireRehire(empId) : UserResponse
UserResponse
success
notRegistered
promoted
demoted
fired
rehired
AccessService
+login(id,password) : UserResponse
EmployeeHandler
+promoteDemote(empId) : UserResponse
+fireRehire(empId) : UserResponse
AccessHandler
+login(id,password) : UserResponse